

DTLS Detailed Explanation

VAI YESKO DETAIL EXPLANATION

2.1.1 Discrete-time linear systems

2.1.1 Discrete-Time Linear Systems (DTLS) – Detailed Explanation

Discrete-time linear systems are one of the most important topics in **Signals and Systems**. Let's break it down step by step in simple English + Nepali mix.

1. What is a Discrete-Time System?

A system that works only at **separate time instants (not continuous time)** is called a discrete-time system.

👉 Input and output are sequences:

- Input: $x[n]$
- Output: $y[n]$

🔗 Example:

- Digital audio processing
- Computer signals
- Mobile communication systems

2. What is a Linear System?

A system is called **linear** if it follows:

(A) Additivity

If:

- $x_1[n] \rightarrow y_1[n]$
- $x_2[n] \rightarrow y_2[n]$

Then:

$$x_1[n] + x_2[n] \rightarrow y_1[n] + y_2[n]$$

(B) Homogeneity (Scaling)

If:

- $x[n] \rightarrow y[n]$

Then:

$$a \cdot x[n] \rightarrow a \cdot y[n]$$

👉 If both conditions are satisfied → system is **linear**.

3. Definition of Discrete-Time Linear System

A system is **discrete-time linear system** if it satisfies:

- ✓ Additivity
- ✓ Homogeneity

👉 Simply:

Input linear combination → Output same linear combination

4. Example of Linear System

System:

$$y[n] = 2x[n]$$

Check:

- If input doubles → output also doubles ✓
- If inputs added → outputs also added ✓

👉 So, this is a **linear system**

5. Example of Non-Linear System

System:

$$y[n] = x^2[n]$$

Check:

- Squaring breaks additivity ✗
- Scaling not preserved ✗

👉 So, this is **non-linear system**

6. Linear Time-Invariant (LTI) System (Important Extension)

Most important type of discrete-time linear systems is:

👉 **LTI System = Linear + Time Invariant**

Time Invariant means:

If input shifts:

$$x[n] \rightarrow y[n]$$

Then:

$$x[n - k] \rightarrow y[n - k]$$

✓ System behavior does not change with time

7. Impulse Response (Very Important)

If input is impulse:

$$\delta[n]$$

Output is:

$$h[n]$$

👉 $h[n]$ is called **Impulse Response**

It fully describes the system.

8. Output of Linear System (Convolution Sum)

For LTI systems:

$$y[n] = x[n] * h[n]$$

Convolution formula:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k]$$

📌 Meaning:

Output = input + system memory combined effect

9. Difference Equation Representation

Discrete-time linear systems are often written as:

$$y[n] + a_1y[n - 1] = b_0x[n] + b_1x[n - 1]$$

👉 This shows:

- Output depends on past outputs
 - Input affects output
-

10. Stability (Extra Important Concept)

A linear system is **stable** if:

- bounded input → bounded output (BIBO stability)

👉 If input is limited, output must also remain limited.

11. Real-Life Example

📱 **WhatsApp voice message system:**

- Input: sound signal $x[n]$
- System: encoding + filtering
- Output: compressed audio signal $y[n]$

If system is linear:

- louder input → proportionally louder output
-

Quick Summary

- ✓ Discrete-time system → works on digital signals
 - ✓ Linear system → satisfies additivity + scaling
 - ✓ LTI system → linear + time invariant
 - ✓ Fully described by impulse response $h[n]$
 - ✓ Output computed using convolution
-

If you want, I can also give:

- ✓ exam questions (MCQ + long questions)
- ✓ diagram/flowchart

✓ numerical convolution examples
just tell 👍

ENGLISH NEPALI MIXUP NO ROMANIZE DETAIL EXPLANATION

2.1.1 Discrete-Time Linear Systems (DTLS) – Detailed Explanation (English + Nepali mix, no romanized Nepali)

Discrete-time linear systems भनेको यस्तो system हो जसले **digital signals** ($x[n]$) लाई **step-by-step discrete time points** मा **process** गर्छ र जसले **linearity (linear behavior)** follow गर्छ।

1. Discrete-Time System के हो?

Discrete-time system भनेको यस्तो system हो जसमा:

- Input signal $\rightarrow x[n]$
- Output signal $\rightarrow y[n]$
- Time continuous हुँदैन, only **separate points** ($n = 0, 1, 2, 3, \dots$) हुन्छ

🔴 Example:

- Mobile phone signal processing
- Computer audio/video processing
- Digital communication systems

👉 यहाँ signal continuous होइन, **sample form** मा हुन्छ

2. Linear System के हो?

A system is called **linear system** if it satisfies two important properties:

(A) Additivity (योग गुण)

यदि:

- $x_1[n] \rightarrow y_1[n]$
- $x_2[n] \rightarrow y_2[n]$

Then:

$$x_1[n] + x_2[n] \rightarrow y_1[n] + y_2[n]$$

🔴 Meaning:

Input add गर्दा output पनि add हुनुपर्छ

👉 नेपाली meaning:

system ले दुई input combine गर्दा output पनि properly combine हुनुपर्छ

(B) Homogeneity (Scaling Property / गुणन नियम)

यदि:

- $x[n] \rightarrow y[n]$

Then:

$$a \cdot x[n] \rightarrow a \cdot y[n]$$

🔴 Meaning:

Input लाई scale गर्दा output पनि same scale मा change हुनुपर्छ

👉 नेपाली meaning:

input बढाए/घटाए अनुसार **output** पनि **proportional change** हुनुपर्छ

3. Discrete-Time Linear System को Definition

A system is linear if:

✓ Additivity hold गर्छ

✓ Homogeneity hold गर्छ

👉 Simple meaning:

Input linear combination भए $\rightarrow$ output पनि same linear combination हुन्छ

4. Example of Linear System (Simple)

System:

$$y[n] = 2x[n]$$

Check:

✓ If input double भयो $\rightarrow$ output पनि double हुन्छ

✓ If inputs add गरियो $\rightarrow$ output पनि add हुन्छ

👉 So, यो system **linear system** हो

🔴 नेपाली meaning:

input लाई **multiply** गर्दा **output** पनि **same rule** follow गर्छ

5. Example of Non-Linear System

System:

$$y[n] = (x[n])^2$$

Problem:

- Input add गर्दा output add हुँदैन ❌
- Scaling गर्दा output proportional हुँदैन ❌

👉 So, यो **non-linear system** हो

👉 नेपाली meaning:

square operation ले **linear rule break** गर्छ

6. Time Invariance (Important Concept)

System लाई time invariant भनिन्छ यदि:

Input shift गर्दा output पनि same shift हुन्छ।

यदि:

$$x[n] \rightarrow y[n]$$

Then:

$$x[n - k] \rightarrow y[n - k]$$

👉 Meaning:

System को behavior समय अनुसार change हुँदैन

👉 नेपाली meaning:

system same तरिकाले काम गर्छ चाहे **input** कहिले आए पनि

7. LTI System (Linear Time Invariant System)

Most important system type:

👉 **LTI = Linear + Time Invariant**

यो system:

✓ Linear हुन्छ

✓ Time invariant हुन्छ

👉 Importance:

👉 Digital signal processing को base यही हो

8. Impulse Response (h[n])

If input is impulse signal:

$$\delta[n]$$

Output हुन्छ:

$$h[n]$$

👉 यसलाई **Impulse Response** भनिन्छ

👉 नेपाली meaning:

system को "fingerprint" हो $h[n]$

👉 यसले system कस्तो छ भनेर पूरा describe गर्छ

9. Output of LTI System (Convolution)

For LTI system:

$$y[n] = x[n] * h[n]$$

Convolution formula:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

👉 Meaning:

Output = input + system effect मिलेर बने result

👉 नेपाली meaning:

input signal लाई system ले filter गरेर output बनाउँछ

10. Difference Equation Representation

Discrete-time linear system लाई equation मा यस्तो लेखिन्छ:

$$y[n] = a_1 y[n-1] + b_0 x[n] + b_1 x[n-1]$$

👉 Meaning:

- Output past output मा depend गर्छ
- Output current input मा पनि depend गर्छ

👉 नेपाली meaning:

system memory भएको जस्तो behavior देखाउँछ

11. Stability (BIBO Stability)

System stable हुन्छ यदि:

👉 bounded input → bounded output

📌 Meaning:

- Input limit मा छ भने output पनि limit भित्र रहनुपर्छ

👉 नेपाली meaning:

system uncontrolled हुनु हुँदैन

12. Real-Life Example (WhatsApp Voice Message)

📱 WhatsApp voice processing:

- Input: sound signal $x[n]$
- System: compression + filtering
- Output: processed audio $y[n]$

👉 If system linear छ भने:

- आवाज double भयो भने output पनि proportional हुन्छ

SHORT SUMMARY

- ✓ Discrete-time system → digital step-by-step signal processing
- ✓ Linear system → additivity + scaling follow गर्छ
- ✓ LTI system → most important practical system
- ✓ Impulse response $h[n]$ → system को identity
- ✓ Convolution → output निकाल्ने main method

If you want, I can also make:

- 📌 diagram / flowchart
- 📌 exam important questions
- 📌 numerical convolution examples
- 📌 MCQs for exam preparation